

Design proposal : Design and Simulation of Unbalanced Mach–Zehnder Interferometer

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Abstract—The main objective of this work is to design and simulate an unbalanced Mach–Zehnder Interferometer (MZI) photonic circuit for waveguide group-index extraction. The path-length difference, ΔL , is varied to investigate its effect on the free spectral range (FSR) around 1550 nm. The simulated FSR values are compared with theoretical values calculated from the waveguide group index.

Index Terms—Mach - Zehnder Interferometer (MZI), Free Spectral Range (FSR)

I. INTRODUCTION

The Mach-Zehnder interferometers (MZIs) are important components in silicon photonic circuits and have applications in optical communications [2], sensing [3], modulation, filtering, and quantum communication. In quantum communication, for example, a 1-bit-delay MZI is used in differential phase-shift quantum key distribution (DPS-QKD) [4]. The output of an MZI is sensitive to the phase difference between its two arms, which results in a periodic transmission spectrum as a function of wavelength.

The aim of this work is to extract the waveguide group index from the measured free spectral range (FSR) of an unbalanced MZI.

II. THEORY

A. Mach-Zehnder interferometer and its transfer function

Consider I_i as the input intensity of the Y branch of an imbalanced MZI, the field equation at the output of the Y branch is given as $E_1 = \frac{E_i}{\sqrt{2}}$ and $E_2 = \frac{E_i}{\sqrt{2}}$. After propagating through the arms, the fields will be $E_{o1} = \frac{E_i}{\sqrt{2}}e^{i\beta_1 L_1}$ and $E_{o2} = \frac{E_i}{\sqrt{2}}e^{i\beta_2 L_2}$, where $\beta_1, \beta_2, L_1, L_2$ are the propagation constants of the two arms and the length of the two arms of the MZI, the propagation constant can be written as

$$\beta(\lambda) = \beta_0 + \beta'_0(\lambda - \lambda_0) + \frac{1}{2}\beta''_0(\lambda - \lambda_0)^2 + \dots \quad (1)$$

For an ideal Y-branch combiner $E_o = \frac{1}{\sqrt{2}}[E_{o1} + E_{o2}]$, neglecting the propagation losses, the output intensity will be;

$$I_o = \frac{I_i}{2} [1 + \cos [\beta_1 L_1 - \beta_2 L_2]] \quad (2)$$

The effective refractive index of both arms are the same, thus $\beta_1 = \beta_2 = \frac{2\pi}{\lambda} n_{eff}$. Thus the output intensity of an imbalanced MZI is dependent on the phase difference between the two arms.

$$\Delta\Phi = \frac{2\pi}{\lambda} n_{eff} [L_1 - L_2] \quad (3)$$

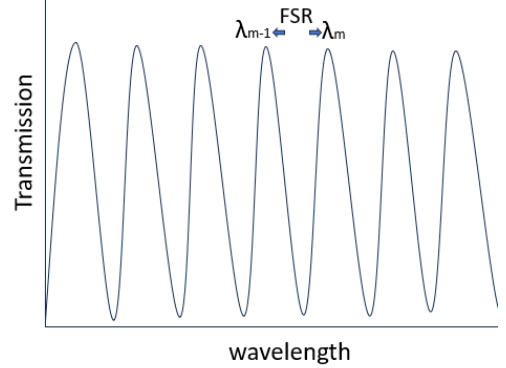


Fig. 1. Schematic transmission spectrum of an MZI, showing the free spectral range (FSR) between adjacent peaks λ_{m-1} and λ_m .

Therefore, the transfer function $T(\lambda)$ can be written [1]

$$T(\lambda) = \frac{I_o}{I_i} = \frac{1}{2} (1 + \cos[\frac{2\pi}{\lambda} n_{eff} \Delta L]) \quad (4)$$

B. Extraction of group index from Free Spectral Range (FSR)

The free spectral range (FSR) can be defined as the separation between two maxima of an optical spectrum. The phase condition for two constructive interferences is shown in figure [1] Let us consider two such maxima at λ_{m-1} and λ_m ,

$$\Delta\Phi_{m-1} = \frac{2\pi}{\lambda_{m-1}} n_{eff} \Delta L = 2(m-1)\pi \quad (5)$$

$$\Delta\Phi_m = \frac{2\pi}{\lambda_m} n_{eff} \Delta L = 2m\pi \quad (6)$$

The difference between $\Delta\Phi_{m-1}$ and $\Delta\Phi_m = 2\pi$.

The phase change can be written as in Taylor series,

$$\Delta\Phi(\lambda) = \Delta\Phi_0 + \Delta\Phi'_0(\lambda - \lambda_0) + \frac{1}{2}\Delta\Phi''_0(\lambda - \lambda_0)^2 + \dots \quad (7)$$

The group refractive index is written as,

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \quad (8)$$

Differentiating $\Delta\Phi(\lambda) = \frac{2\pi}{\lambda} n_{eff} \Delta L$ gives

$$\frac{d\Delta\Phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} \left(n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \right) = -\frac{2\pi\Delta L n_g}{\lambda^2}. \quad (9)$$

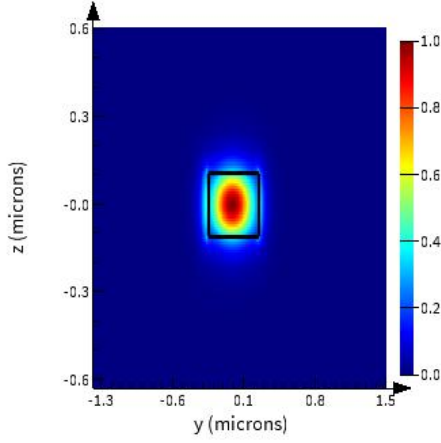


Fig. 2. Lumerical FDE simulated quasi-TE mode profile of the silicon waveguide at 1550 nm.

Between two adjacent maxima the phase changes by 2π , so

$$\left| \frac{d\Delta\Phi}{d\lambda} \right| \Delta\lambda = \frac{2\pi\Delta L n_g}{\lambda_0^2} \Delta\lambda = 2\pi, \quad (10)$$

which is Eq. (10). Solving for n_g gives

$$n_g = \frac{\lambda_0^2}{\Delta L \Delta\lambda}, \quad (11)$$

where $\Delta\lambda$ is the free spectral range (FSR).

III. MODELING AND SIMULATIONS

A. Waveguide modelling

TABLE I
WAVEGUIDE PARAMETERS USED IN THE SIMULATION.

Parameter	Value
Waveguide material	Silicon (Si)
Cladding material	Silicon dioxide (SiO ₂)
Waveguide width	0.500 μm
Waveguide height	0.220 μm
Polarization	Quasi-TE

The mode profile is simulated at $1.55\mu\text{m}$ and the profile is shown in the figure 2. The effective index at $1.55\mu\text{m}$ is $n_{\text{eff}} = 2.444353$ and group index n_g is 4.207933.

In Lumerical MODE, there's an inbuilt frequency analysis from which the variation of effective refractive index with wavelength and variation of group index with wavelength is determined across $1.5\mu\text{m} - 1.6\mu\text{m}$ figure 3 and figure 4 respectively.

For the circuit-level analysis in Lumerical INTERCONNECT, the wavelength dependence of the effective index was approximated by a quadratic model about $\lambda_0 = 1.55 \mu\text{m}$ (wavelength in μm):

$$n_{\text{eff}}(\lambda) = 2.444353 - 1.137818(\lambda - 1.55) - 0.0144963(\lambda - 1.55)^2, \quad (12)$$

with coefficients fitted to Lumerical FDE data over 1.50–1.60 μm . The FDE calculation gives $n_{\text{eff}} = 2.444353$ and a group index $n_g = 4.207933$ at 1.55 μm .

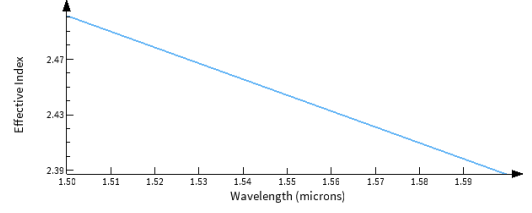


Fig. 3. Simulated effective index for quasi-TE mode from $1.5\mu\text{m} - 1.6\mu\text{m}$

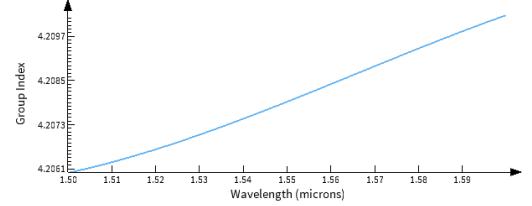


Fig. 4. Simulated group index for quasi-TE mode from $1.5\mu\text{m} - 1.6\mu\text{m}$

B. MZI simulation

The dependence of the free spectral range (FSR) on the path-length difference was studied using an unbalanced Mach-Zehnder interferometer (MZI) in Lumerical INTERCONNECT (Fig. 5). The shorter arm was fixed at $L_2 = 50 \mu\text{m}$, and the longer arm L_1 was varied from 80 to 185 μm in 15 μm steps, giving $\Delta L = 30-135 \mu\text{m}$.

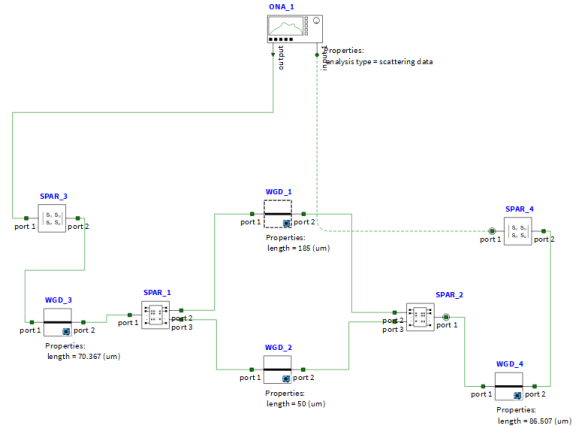


Fig. 5. MZI circuit in Lumerical INTERCONNECT.

C. FSR-based group-index extraction

For each design, the FSR was read from the simulated transmission spectrum at the peak nearest to 1550 nm (1546–1552 nm for $\Delta L \geq 45 \mu\text{m}$). The simulated FSR was compared with the analytical value $\text{FSR}_{\text{calc}} = \lambda^2 / (n_g \Delta L)$, using the FDE group index $n_g = 4.207933$. The variation is

$$\text{Variation} = \frac{|\text{FSR}_{\text{sim}} - \text{FSR}_{\text{calc}}|}{\text{FSR}_{\text{calc}}} \times 100\%. \quad (13)$$

The group index was then extracted from each simulated FSR as $n_g = \lambda^2 / (\text{FSR}_{\text{sim}} \Delta L)$ (Table II).

TABLE II
CALCULATED AND SIMULATED FSR, VARIATION, AND EXTRACTED GROUP INDEX FOR DIFFERENT MZI DESIGNS ($L_2 = 50 \mu\text{m}$).

L_1 (μm)	ΔL (μm)	FSR_{calc} (nm)	FSR_{sim} (nm)	Variation (%)	Extracted n_g
80	30	19.032	18.930	0.54	4.229
95	45	12.688	12.802	0.90	4.169
110	60	9.516	9.308	2.18	4.281
125	75	7.613	7.391	2.91	4.345
140	90	6.344	6.278	1.04	4.250
155	105	5.438	5.272	3.05	4.327
170	120	4.758	4.697	1.27	4.268
185	135	4.229	4.290	1.44	4.146

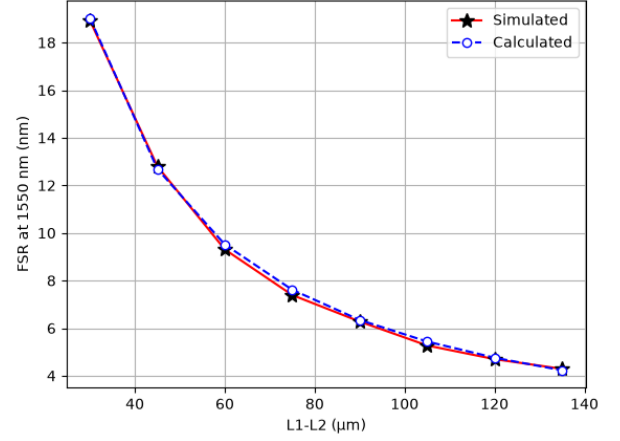


Fig. 7. Analytical and INTERCONNECT-simulated FSR as a function of ΔL .

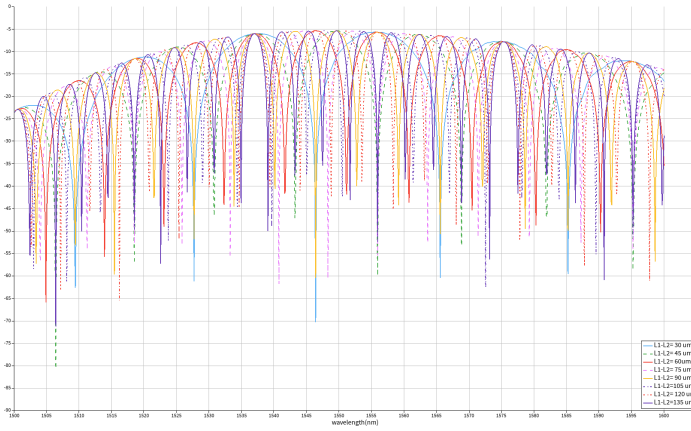


Fig. 6. Optical spectra of unbalanced MZIs with different ΔL (500 nm $\times$ 220 nm strip waveguide, TE).

As expected, the FSR decreases with increasing ΔL , and the simulated and calculated values agree within 3.1% for all eight cases (Fig. 7). The extracted group indices range from 4.146 to 4.345, with an average of 4.252, about 1% above the FDE value of 4.208.

D. Layout for Fabrication

E. Layout

The mask layout (Fig. 9) was drawn in KLayout using SiEPIC-Tools with the SiEPIC EBeam PDK. It contains 8 unbalanced MZIs with ΔL ranging from 30 to 135 μm in 15 μm steps, all built from the same 500 nm $\times$ 220 nm strip waveguide, Y-branches and grating couplers (GCs). Each MZI is connected to an input and an output GC placed [127 μm] apart so that it can be probed with a standard fibre array. The shorter arm was fixed at $L_2 = 50 \mu\text{m}$ in every design, and only L_1 was changed. Each device carries a text label on its input GC for identification during measurement [e.g. `opt_in_TE_1550_device_gourisankar_MZI(id from 1-9)`].

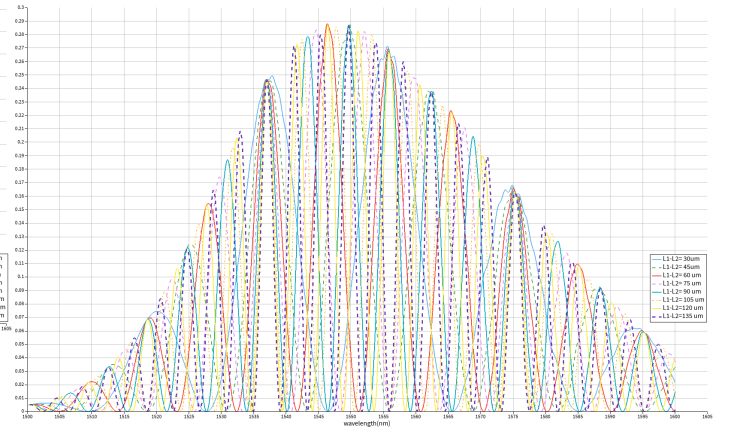


Fig. 8. Transmission spectra of unbalanced MZIs with different ΔL (500 nm $\times$ 220 nm strip waveguide, TE).

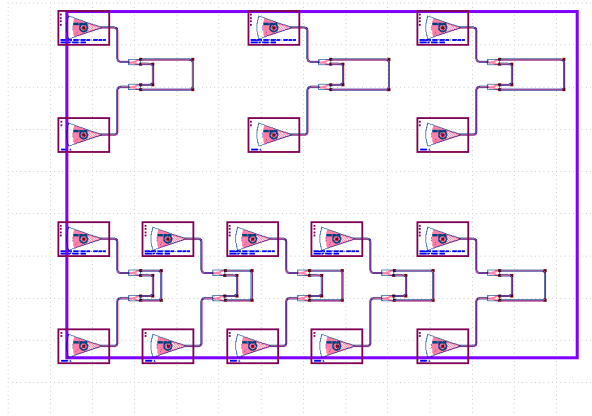


Fig. 9. Layout of the [605 $\times$ 410] μm cell showing the unbalanced MZIs and their grating couplers.

F. Layout verification

G. Layout verification

The layout was verified in SiEPIC-Tools (v0.5.31) in KLayout in two steps. First, the KLayout design rule check (DRC) with the SiEPIC-EBeam-PDK rule deck was run on the final GDS. It covers the boundary, Si, SiN and metal width and spacing rules, and it returned no violations in any category. Second, the SiEPIC functional layout check, which verifies waveguide connectivity and the match between components and the extracted netlist, was run and reported no layout errors. Together these confirm that the layout is free of design-rule and connectivity errors before the circuit netlist was extracted for simulation.

H. Simulation of the layout

The circuit netlist was extracted from the final layout and simulated in Lumerical INTERCONNECT using the PDK compact model library, so that the results include the layout's actual arm lengths, bends and couplers rather than the ideal schematic values. The wavelength was swept from 1500 to 1600 nm with 2000 points. From each spectrum, the FSR was read at the peak closest to 1550 nm, the group index was calculated from it as $n_g = \lambda^2 / (\text{FSR} \Delta L)$. The extracted values in Table III.

TABLE III
FSR AND EXTRACTED GROUP INDEX OF THE SIMULATED LAYOUT MZIS
($L_2 = 50 \mu\text{m}$).

L_1 (μm)	ΔL (μm)	FSR (nm)	n_g
80	30	18.154	4.411
95	45	12.808	4.169
110	60	9.412	4.254
125	75	7.398	4.330
140	90	6.182	4.318
155	105	5.6123	4.007
170	120	4.800	4.171
185	135	4.293	4.145

IV. REFERENCES

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2015.
- [2] Ran, S., Guo, Y., Liu, Y. et al. A 4×256 Gbps silicon transmitter with on-chip adaptive dispersion compensation. *Nat Commun* 16, 6268 (2025). <https://doi.org/10.1038/s41467-025-61408-7>.
- [3] Kazanskiy, N.L.; Khonina, S.N.; Butt, M.A. Advancement in Silicon Integrated Photonics Technologies for Sensing Applications in Near-Infrared and Mid-Infrared Region: A Review. *Photonics* 2022, 9, 331. <https://doi.org/10.3390/photonics9050331>.
- [4] A. Atkar, H. Saraf, V. Kumar, A. Singh and P. Sharma, "PIC based Implementation of DPS-QKD Transceiver," 2025 International Conference on Microwave, Optical, and Communication Engineering (ICMOCE), Bhubaneswar, India, 2025, pp. 1-5, doi: 10.1109/ICMOCE64100.2025.11076980.